The uniformity problem is that the standard Big Bang cosmology doesn't offer any explanation for how the whole observable universe, including parts on totally opposite sides of the sky, ended up being the same temperature at early times. When we look at the afterglow of the Big Bang, we see it as the same everywhere to extremely high precision, which is, when you think about it, a really strange coincidence. Normally, two things can become the same temperature if they're in what we call *thermodynamic equilibrium*. This just means there's a way for them to exchange heat, and time for them to do so. If you leave a cup of coffee in a room for long enough, the coffee and the air in the room will interact, and eventually you'll have room-temperature coffee and a very slightly warmer room. The problem with the standard picture of the early universe is that it doesn't include a situation in which two distant parts of the universe could interact and come to an agreement about a temperature. If we take two points on opposite sides of the sky and work out how far apart they are now, and how far apart they were at the very beginning, 13.8 billion years ago, it becomes clear that there was never a time in the history of the universe when they were close enough that light beams could have traveled back and forth between them to carry out the equilibrium process. A beam of light that started at one of those points at the beginning of the universe would not have had time even in 13.8 billion years to cover the distance necessary to get to the other. They are, and always have been, outside each other's horizons: unable to communicate in any way.[fn14] So either it's the most massive coincidence in the universe, or something happened early on to let the equilibrium happen.

The problem with the imperfections is a little simpler to articulate. It's just this question: where did those minuscule density fluctuations in the cosmic microwave background come from, and why are they patterned the way they are?

Cosmic inflation solves both these problems, along with a few others. The basic idea is that there was a time in the early universe, after the singularity but before the end of the primordial fireball stage, when the universe was expanding astonishingly fast. This helps by allowing for a period early on when a very small region could come into equilibrium, after which the rapid expansion would take that little nicely settled region and stretch it out to cover our entire observable universe. Imagine taking a complicated abstract painting and blowing it up so large that your whole